shows a Python script being reported as suspicious. These outputs can be captured to a text file log, or parsed with a *grep* command to remove the clutter and only concentrate on important messages. Using –q is possible, but it can suppress suspect items, which may not be a good idea.

What if a root-kit is found?

Chkrootkit detects root-kits, but does not remove those. Upon finding a root-kit on a system, the first thing to do is to remove it from the network, to avoid its further spread. To remove a root-kit in the cleanest way, you need to back up and rebuild the entire system, which may not always be possible. Another approach is to study the detected root-kit thoroughly, and perform actions to remove it based on its method of intrusion and how it works. Many root-kits can be removed manually; however, there are a few which need only the cleanest approach. To improve detection accuracy, it is advisable that you run *Chkrootkit* from a known healthy system, against all servers in the farm.

Integrating *Chkrootkit* for admin tasks

Systems administrators are strongly advised to use *Chkrootkit* in their daily administrative tasks. By using an appropriate command-line option, you can create a script to have verbose output, and dump it into a log file which can be further parsed to look for anomalies. This script can be set (using a *cron* job) to run on a daily schedule. Over time, the script can be tuned further to remove false alarms generated by suspicious files and report only real problems. Another *cron* job can be scheduled more frequently, to have *Chkrootkit* detect if network interfaces are in promiscuous mode. This is essential because, usually, a root-kit attack starts by tampering with the network configuration. Further scripting can be done to create a list of servers in the farm that need to be scanned, their outputs can be consolidated, and alerts or reports can be made to administrators accordingly. END

Summary

Root-kits are a serious threat to modern data centres. It is essential for the IT management team to have definite means to detect them and take responsive action. *Chkrootkit* is a fast and effective scanning tool for just this purpose. Its command line options help to automate runs on a periodic basis, which is good for administrators.

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